

24.1 A 1Mb Multibit ReRAM Computing-In-Memory Macro with 14.6ns Parallel MAC Computing Time for CNN-Based AI Edge Processors

Cheng-Xin Xue, Wei-Hao Chen, Je-Syu Liu, Jia-Fang Li, Wei-Yu Lin, Wei-En Lin, Jing-Hong Wang, Wei-Chen Wei, Ting-Wei Chang, Tung-Cheng Chang, Tsung-Yuan Huang, Hui-Yao Kao, Shih-Ying Wei, Yen-Cheng Chiu, Chun-Ying Lee, Chung-Chuan Lo, Ya-Chin King, Chong-Jung Lin, Ren-Shuo Liu, Chih-Cheng Hsieh, Kea-Tiong Tang, Meng-Fan Chang

National Tsing Hua University, Hsinchu, Taiwan

Embedded nonvolatile memory (NVM) and computing-in-memory (CIM) are significantly reducing the latency (t_{MAC}) and energy consumption (E_{MAC}) of multiply-and-accumulate (MAC) operations in artificial intelligence (AI) edge devices [1,2]. Previous ReRAM CIM macros demonstrated MAC operations for 1b-input, ternary-weighted, 3b-output CNNs [1] or 1b-input, 8b-weighted, 1b-output fully-connected networks with limited accuracy [2]. To support higher-accuracy convolution neural network heavy applications NVM-CIM should support multibit inputs/weights and multi-bit output (MAC-OUT) for CNN operations. One way to achieve multibit weights is to use a multi-level ReRAM cell to store the weight. However, as shown in Fig. 24.1.1, multibit ReRAM CIM faces several challenges. (1) a tradeoff between area and speed for multibit input/weight/MAC-OUT MAC operations; (2) sense amplifier's high input offset, large area, and high parasitic load on the read-path due to large BL currents (I_{BL}) from multibit MAC; (3) limited accuracy due to a small read/sensing margin (I_{SM}) across MAC-OUT or variation in cell resistance (particularly MLC cells). To overcome these challenges, this work proposes, (1) a serial-input non-weighted product (SINWP) structure to optimize the tradeoff between area, t_{MAC} and E_{MAC} ; (2) a down-scaling weighted current translator (DSWCT) and positive-negative current-subtractor (PN-ISUB) for short delay, a small offset and a compact read-path area; and (3) a triple-margin small-offset current-mode sense amplifier (TMCSA) to tolerate a small I_{SM} . A fabricated 55nm 1Mb ReRAM-CIM macro is the first ReRAM CIM macro to support CNN operations using multibit input/weight MAC-OUT. This device achieves the shortest CIM-MAC-access time (t_{AC}) among existing ReRAM-CIMs ($t_{MAC}=14.6ns$ with 2b-input, 3b-weight with 4b-MAC-OUT) and the best peak E_{MAC} of 53.17 TOPS/W (in binary mode).

Figure 24.1.2 presents the structure of the proposed ReRAM-CIM macro, using a 1T1R SLC ReRAM array, a current-aware BL clumper (CABLC), DSWCT, SINWP sampler-and-combiner (SINWP-SC), PN-ISUB, and TMCSA. This work places both positive and negative weights in the same array, but in different columns, unlike [1]. Each 3b-signed weight ($W=1b\text{-sign}+2b\text{-data}$) is stored in the same row of either the 2-column positive (PWG) or negative (NWG) group. In each PWG/NWG, the even BLs (BLM) represent the MSB and the odd BLs (BLL) represent the LSB. In other words, 4 SLC ReRAM cells are used to store 3b signed weights. To support configurable 1b/2b input (IN), this work translates 2b-input into two sequential single-bit (IN_0 and IN_1) WL pulses in one CIM clock cycle. For an n -by- n CNN kernel, n^2 weights are stored in n^2 consecutive rows. CABLC clamps the BL voltage (V_{BLC}) for current-mode sensing. A BL current (I_{BL}) that is equal to the sum of n^2 cell currents (I_{MC}) is generated via binary multiplication ($IN \times W$) between a WL (IN) and a 1b-weight (W), as in [1]. The I_{BL} of BLM (I_{BL-MSB}) and BLN (I_{BL-LSB}) do not include their place-value processed in array (non-weighted I_{BL}). DSWCT and SINWP-SC then combine the I_{BL-MSB} and I_{BL-LSB} of a PWG/NWG to generate a weighted data-line (DL) current (I_{DL}) for the MAC value (MACV) using 2b-inputs and 2b-weights. PN-ISUB outputs the difference (I_{SUB}) between the I_{DL} of PWG (I_{DL-P}) and NWG (I_{DL-N}) as well as the sign bit (DOUT_{SIGN}=0 when $I_{DL-P} > I_{DL-N}$), where $I_{SUB}=|I_{DL-P}-I_{DL-N}|$. The current reduction provided by DSWCT and PN-ISUB allows for an I_{SUB} that is much smaller than the total current ($I_{DL-MSB}+I_{DL-LSB}$) of the accessed PWG and NWG. This allows TMCSA to use smaller transistors (achieving a lower parasitic load), which results in faster response times, smaller area and less power compared to schemes without DSWCT and PN-ISUB. Finally, each IO repeats the operation of TMCSA in 3 sequential phases to detect the same I_{SUB} with 3 different I_{REF} currents (I_{REF1} to I_{REF3}), and then outputs a 4b MACV with 3b data from TMCSA and a sign bit from PN-ISUB.

Figure 24.1.3 shows the operations of DSWCT, SINWP-SC, and PN-ISUB. DSWCT uses a current-mirror (CM) to translate I_{DL-MSB} and I_{DL-LSB} into lower weighted DL currents (I_{WDL}). $I_{WDL-MSB}=(1/p) \cdot 2 \cdot I_{DL-MSB}$ and $I_{WDL-LSB}=(1/p) \cdot I_{DL-LSB}$. Choice of the reduced-amplitude ratio, p ($p=4$ for this work) is based on maintaining sufficient I_{SM} , while reducing power consumption and the transistor size (area and parasitic load) required for the remaining read-path. SINWP-SC operations occur in two phases. (1) For the 1st WL pulse (IN_0), $IN_{SW}=1$ and the gate-voltage ($V_{GCM-MSB}/V_{GCM-LSB}$) of

CM_{MSB}/CM_{LSB} of DSWCT is sent to N3/N6 and stored in capacitor CM/CL to sample the $I_{WDL-MSB}/I_{WDL-LSB}$ resulting from $IN_0 \times W$. (2) For the 2nd WL pulse (IN_1), $IN_{SW}=0$ and $V_{GCM-MSB}/V_{GCM-LSB}$ is sent to N4/N7. The size of N4/N7 is 2x that of N3/N6. SINWP-SC then combines the drain currents of N3, N4, N6 and N7 to output a weighted I_{DL} as follows:

$$I_{DL-LSB[0]} \cdot W_{DL-LSB[0]}/8 + I_{DL-MSB[0]} \cdot W_{DL-MSB[1]}/4 + I_{DL-LSB[1]} \cdot W_{DL-LSB[0]}/4 + I_{DL-MSB[1]} \cdot W_{DL-MSB[1]}/2$$

In PN-ISUB, a comparator compares I_{DL-P} with I_{DL-N} to output the sign bit (DOUT_{SIGN}=0 when $I_{DL-P} > I_{DL-N}$), and DOUT_{SIGN} enables PN-ISUB to connect the larger/lower current path of DL to the high/low-current input (I_{HC}/I_{LC}) terminal of a current-subtractor. If DOUT_{SIGN} = 1, $I_{HC}=I_{DL-N}$, and $I_{LC}=I_{DL-P}$, then the current-subtractor generates the PN-ISUB output current $I_{SUB}(=I_{HC}-I_{LC}=I_{DL-N}-I_{DL-P}$ if DOUT_{SIGN} = 1). Compared to schemes that convert I_{DL-P}/I_{DL-N} to MACV separately before the digital combiner as in [1], PN-ISUB removes current leakage from HRS cells ($n^2 \cdot I_{HRS}$) when $I_{DL-N} \gg I_{DL-P}$, or enable larger current reduction (or small I_{SUB}) when $I_{DL-N} \sim I_{DL-P}$ to improve sensing yield in the CSA.

Figure 24.1.4 shows the operation of TMCSA, which comprises two pairs of PMOS transistors (P1:P2 and P3:P4), four pairs of switches (SW1-SW4), two overdrive-coupling capacitors (C1, C2), four discharge NMOSs (DN1-DN4), an NMOS latch (N1-N3) and two current inputs (I_{IN} and I_{REF}). In stand-by mode, SW3 and SW4 are on, and SW1, SW2, and VDD_{SA} are off. DSD and CHD are high and DN1-DN4 are turned on to hold nodes DP1, DP2, LQ and LQB at OV. In phase-1 (PH1, V_{TH} sampling), DSD, SW1 and SW2 are off and VDD_{SA} is on. The threshold voltages (V_{TH1} to V_{TH4}) of diode-connected P1 to P4 are stored on their gates (i.e. $V_{G1}=V_{DD}-V_{TH1}$, $V_{G3}=V_{DD}-V_{TH3}$). In phase-2 (PH2, V_{OV} sampling and coupling), SW4 is off, SW1 is on, and I_{IN}/I_{REF} flows through P1/P2. This results in $V_{G1}=V_{DD}-V_{TH1}-V_{OV-IN}$ and $V_{G2}=V_{DD}-V_{TH2}-V_{OV-REF}$, where V_{OV-IN}/V_{OV-REF} is the overdrive voltage of P1/P2 for sampling I_{IN}/I_{REF} as in [3]. In the meantime, V_{OV1}/V_{OV2} is coupled to V_{G4}/V_{G3} via C1/C2, so that $V_{G4}=V_{DD}-V_{TH4}-V_{OV1}$ and $V_{G3}=V_{DD}-V_{TH3}-V_{OV2}$. Ideally, $I_{P4}=2 \cdot I_{IN}$ and $I_{P3}=2 \cdot I_{REF}$ since transistors P3 and P4 are twice as big (2-finger style) as P1 and P2. In phase-3 (PH3, ΔI amplifying), SW2 is on and SW1 and SW3 are off. Thus, the current at node LQ is $I_{P3}-I_{IN}=2 \cdot I_{REF}-I_{IN}$. The current at node LQB is $I_{P4}-I_{REF}=2 \cdot I_{IN}-I_{REF}$. For a given period (T_{PH3}), the voltage difference (V_{LQ-LQB}) between node LQ and LQB is proportional to $[(2 \cdot I_{REF}-I_{IN})-(2 \cdot I_{IN}-I_{REF})] \cdot T_{PH3}=3 \cdot (I_{IN}-I_{REF}) \cdot T_{PH3}$, which is 3x the $I_{SM}(=I_{IN}-I_{REF})$ in conventional CSA. In phase-4 (PH4, latch), SAEN is high and N1-N3 are enabled to detect V_{LQ-LQB} and generate a digital output at SAOUT.

Figure 24.1.5 shows the performance of the proposed schemes. For 3x3-CNN kernels DSWCT reduces the worst-case current by 3.6x. The combination of DSWCT and PN-ISUB schemes enables a 3.6–3.8x reduction in MACV-current. The SINWP scheme achieves an FoM ($SM/(Energy \times Area)$) 1.4–6x better than those of sequential-input-parallel-weight (SIPW) and parallel-input-parallel-weight (PIPW) structures. TMCSA enabled 6x and 1.7x reductions in input offset, compared to conventional CSA and DR-CSA [1], using a 70pA input current.

Figure 24.1.6 presents the measurement results from a 1Mb ReRAM-CIM macro fabricated using 1T1R SLC ReRAM in a 55nm CMOS process. A demo system was built using our ReRAM-CIM with an FPGA host. For CNN operations using 3x3 kernels and 2b-input and 3b-weight, the captured waveforms confirm that t_{MAC} for a 4b MAC output (1b sign and 3b data) is 14.6ns, excluding the path-delay. Using 3x3-CNN kernels with 1b-input and 3b-weight, shmoo results confirm a t_{MAC} of 11.75ns per cycle at typical V_{DD} , and the system test achieve an 88.52% inference accuracy on the CIFAR-10 dataset. The measured peak energy-efficiency is 53.17TOPS/W in binary mode (1b-input, 3b-weight, 4b-MAC-OUT) and 21.9TOPS/W in multibit mode (2b-IN, 3b-weight, 4b-MAC-OUT) using CIM peripheral circuits and a reference generator. This work also achieves a 3.4x improvement in energy-efficiency and a 1.3x faster t_{MAC} , compared to [1]. Figure 24.1.7 presents the die micrograph and chip summary.

Acknowledgements:

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References:

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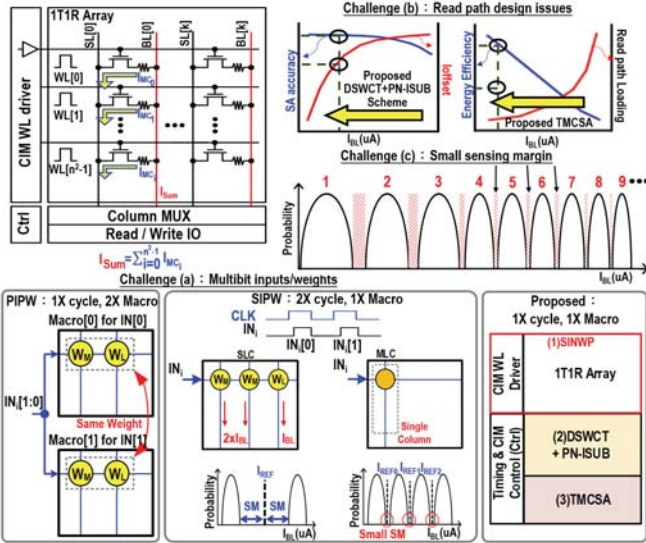


Figure 24.1.1: Multi-bit computation in nonvolatile memory.

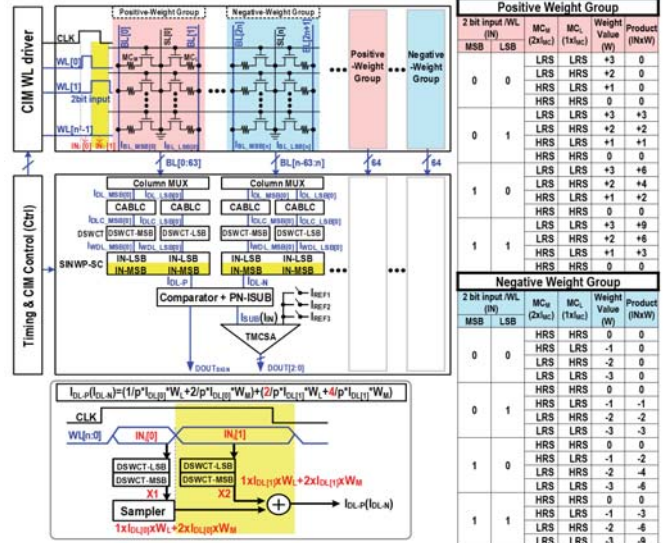


Figure 24.1.2: Proposed ReRAM-CIM macro.

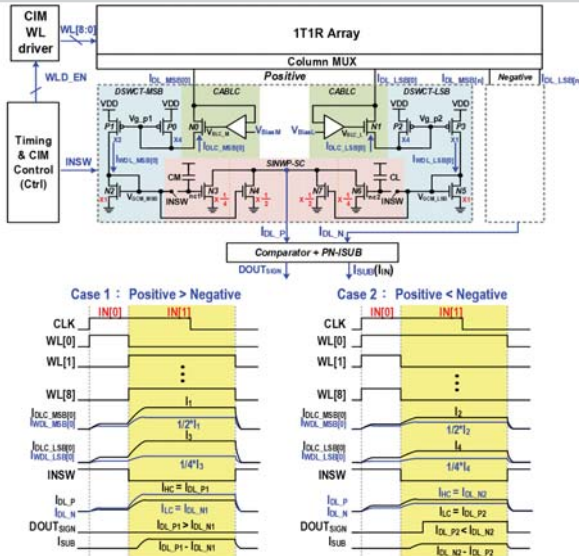


Figure 24.1.3: Sequential input parallel weight structure.

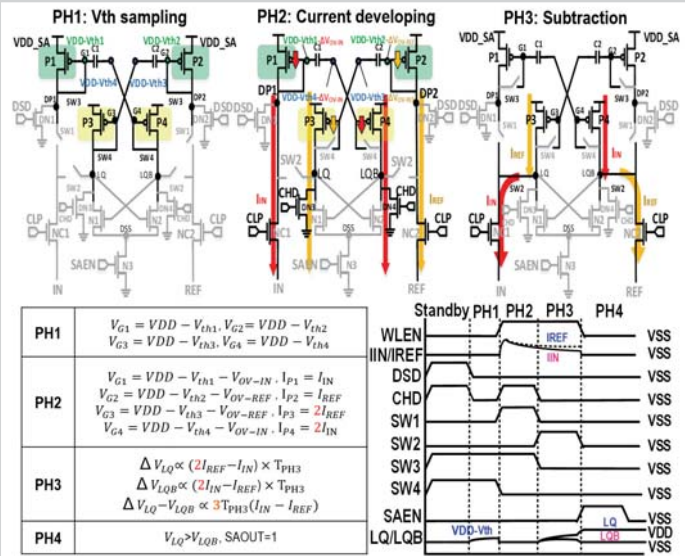


Figure 24.1.4: Structure and operation of TMCSA.

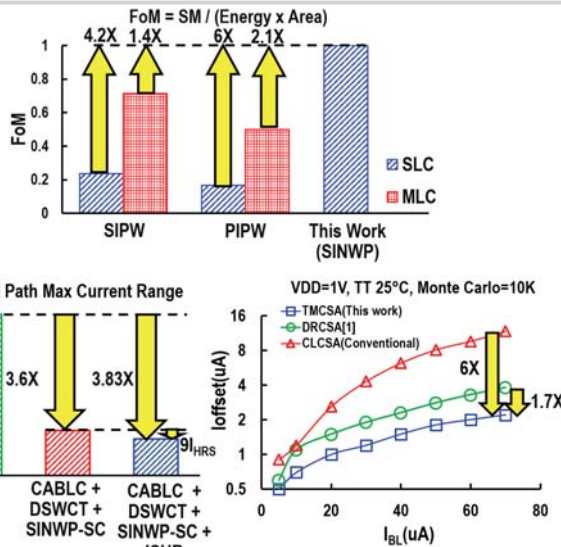


Figure 24.1.5: Performance comparison of this work vs prior-art.

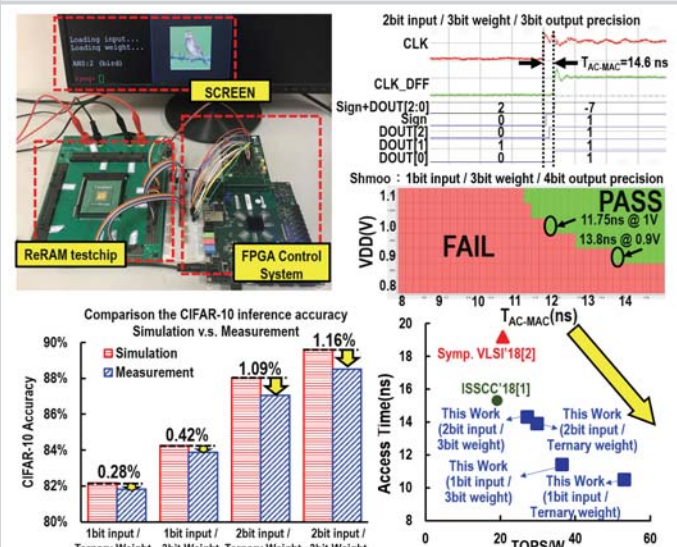
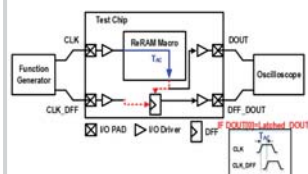


Figure 24.1.6: Measurement results.



Technology	55nm CMOS Logic Process
ReRAM	Logic-process ReRAM
Cell Size(1T1R)	0.2025 μm^2
ReRAM Mode	Memory/CIM
CIM Mode	CNN
Capacity	1Mb (8 Sub-array)
Sub-array	256rows x 512columns
Read Delay @ VDD =1V	Memory Mode: 3.16ns
	CNN Mode: 11.75ns (1bit input / 3bit weight) (3bit DOUT precision)
	CNN Mode: 14.6ns (2bit input / 3bit weight) (3bit DOUT precision)
Energy Efficiency @ VDD =1V	CNN Mode: 53.17 (TOPS/W) (1bit input / 3bit weight) (3bit DOUT precision)
	CNN Mode: 21.9 (TOPS/W) (2bit input / 3bit weight) (3bit DOUT precision)

Figure 24.1.7: Die micrography and summary Table.